A Variable-Mass Alternative to Dark Matter and Galactic Rotation Curves

by

Jonah Cooke

A thesis submitted to the

Faculty of the Graduate School of the

University of Colorado in partial fulfillment

of the requirements for the degree of

Bachelor of Science

Department of Physics

2026

Committee Members:

TBD, Chair

TBD

Cooke, Jonah (B.S., Physics)

A Variable-Mass Alternative to Dark Matter and Galactic Rotation Curves

Thesis directed by Professor TBD

Dedication

For whom it may concern.

Acknowledgements

Thanks to everyone who helped.

Contents

Tables

Table

Figures

Chapter 1

Introduction

Dark matter has evaded detection efforts since it's prediction, in the eighties [1–3], indicating the potential need for an alternative. In this paper, I explore the effect of variable mass (variable higgs field vacuum expectation) on galactic rotation curves. This study is based off findings of universal accelleration, and spectra shifts due to spatially dependant mass from [3].

Chapter 2

Review of literature

2.1 General Relativity

2.2 Dark Matter

2.3 Higgs field

Chapter 3

Methods

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

References

[1] V. C. Rubin, W. K. Ford Jr., and N. Thonnard, "Rotational properties of 21 SC galaxies with a large range of luminosities and radii, from NGC 4605 (R=4kpc) to UGC 2885 (R=122kpc).", The Astrophysical Journal **238**, 471 (1980) (Cited on p. 1).

[2] V. C. Rubin, "EXTENDED ROTATION CURVES OF HIGH-LUMINOSITY SPIRAL GALAXIES. IV. SYSTEMATIC DYNAMICAL PROPERTIES, Sa^Sc", ApJ. . . (Cited on p. 1).

[3] Y. Shi, *Force, metric, or mass: Disambiguating causes of uniform gravity*, (2021) http://arxiv.org/abs/1908.02159 (visited on 04/14/2026), pre-published (Cited on p. 1).